TRAFFIC DEMAND PREDECTION USING MACHINE LEARNING

Problem statement:

Accurately predict traffic demand(vehicle count)for future time intervals at different road segments considering weather,events,time and road infrastructure

Contents:

Abstract

Traffic congestion has become a major challenge in urban areas due to the increasing number of vehicles. Predicting traffic demand in advance helps improve traffic management and reduces congestion.

This project develops a **Traffic Demand Prediction System** using Machine Learning. A synthetic dataset was generated using Python with features such as road type, number of lanes, weather conditions, temperature, humidity, speed limit, nearby intersections, and event type. Data preprocessing and feature selection were performed before training the machine learning models.

Two regression algorithms, **LightGBM** and **XGBoost**, were trained and evaluated using R^2 Score, MAE, and RMSE. The best-performing model was saved and integrated into a Flask web application where users can enter traffic-related details and receive predicted traffic demand.

Introduction

Traffic congestion affects travel time, fuel consumption, and air pollution. Machine Learning helps analyze historical traffic patterns and predict future traffic demand. This project provides a simple web-based application that predicts traffic demand using road, weather, and time-related information.

Objectives

- Generate a synthetic traffic dataset.
- Preprocess and clean the dataset.
- Select important features.
- Train LightGBM and XGBoost models.
- Compare model performance.
- Develop a Flask web application.
- Predict traffic demand based on user inputs.

Dataset & Features

Dataset Description

A synthetic traffic dataset was generated using Python.

The dataset contains:

- Road Type
- Number of Lanes
- Speed Limit
- Day of Week
- Hour
- Weather Condition
- Temperature
- Humidity

- Rainfall
- Wind Speed
- Visibility
- Nearby Intersections
- Event Type
- Peak Hour
- Traffic Demand (Target Variable)

Methodology

Dataset Generation

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Data Preprocessing

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Feature Selection

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Train-Test Split

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LightGBM Model

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XGBoost Model

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Model Comparison

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Best Model Selection

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Flask Web Application

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Traffic Prediction

Machine Learning Models

LightGBM

Models Used

LightGBM is a gradient boosting algorithm designed for fast and accurate predictions. It efficiently handles large datasets and improves prediction performance.

XGBoost

XGBoost is an optimized gradient boosting algorithm widely used for regression and classification tasks. It provides high prediction accuracy and handles complex relationships between features.

Evaluation Metrics

- R^2 Score
- Mean Absolute Error (MAE)
- Root Mean Square Error (RMSE)

Output:

```
===== LightGBM Results =====
R² Score : 0.9488
MAE      : 10.6291
RMSE     : 13.6827

===== XGBoost Results =====
R² Score : 0.9488
MAE      : 10.6276
RMSE     : 13.6874

Best Model: LightGBM
Model saved successfully as traffic_model.pkl
```

Results:

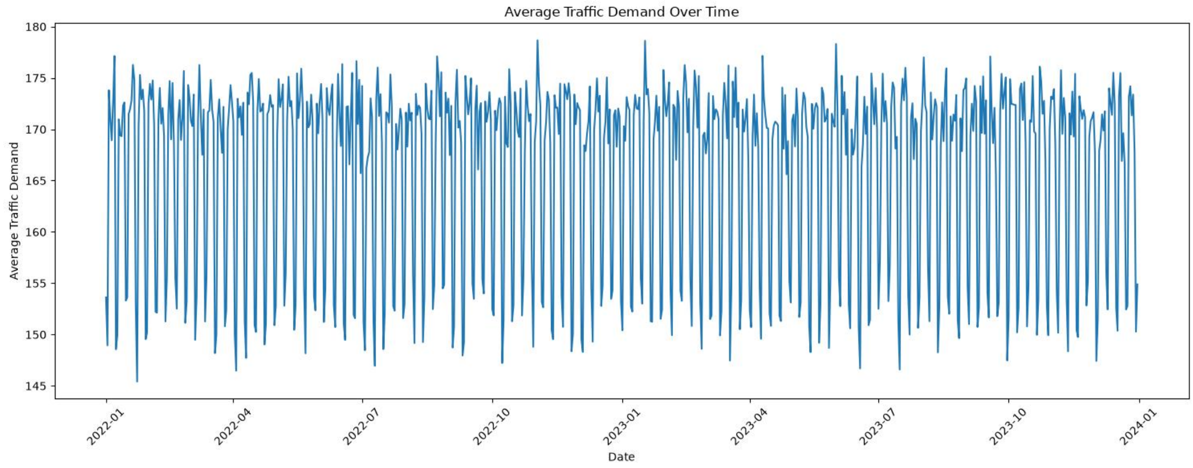

Fig1:Traffic Demand Over Time

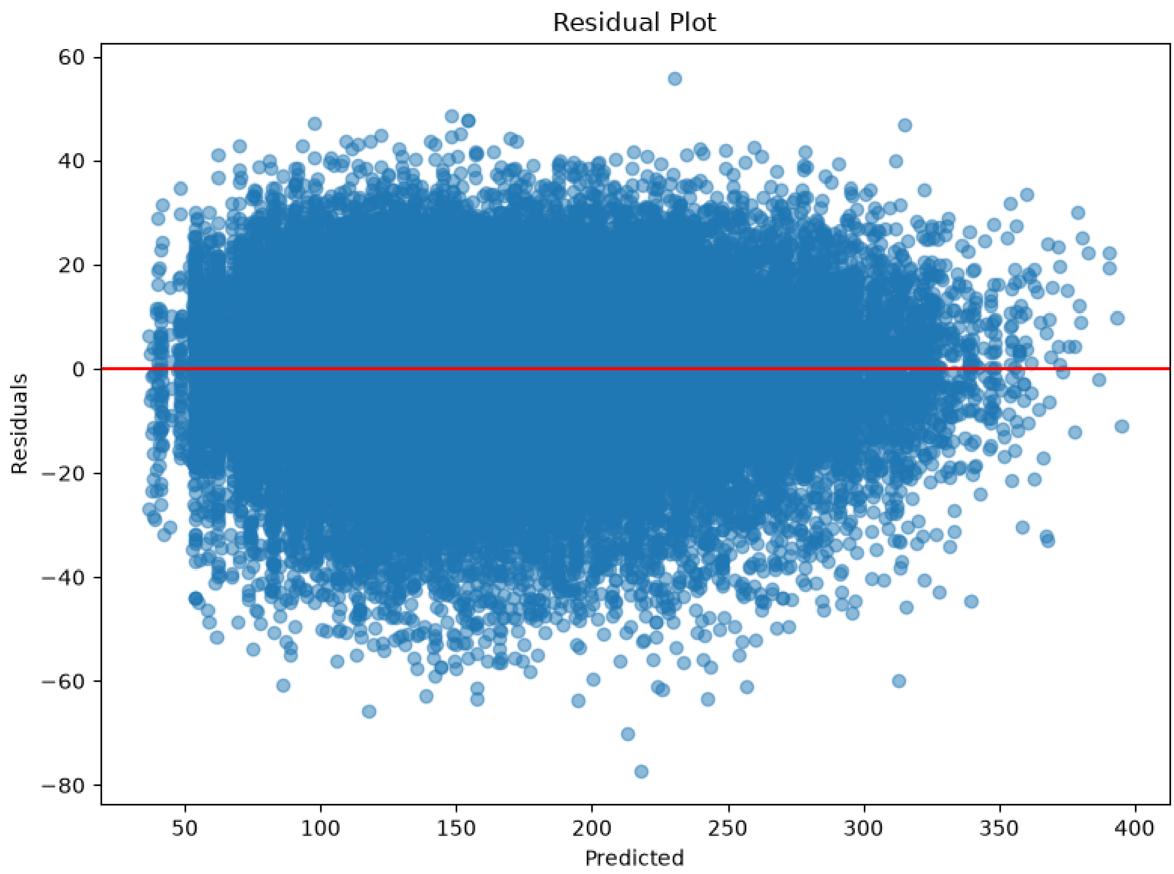

Fig2:Residual Plot

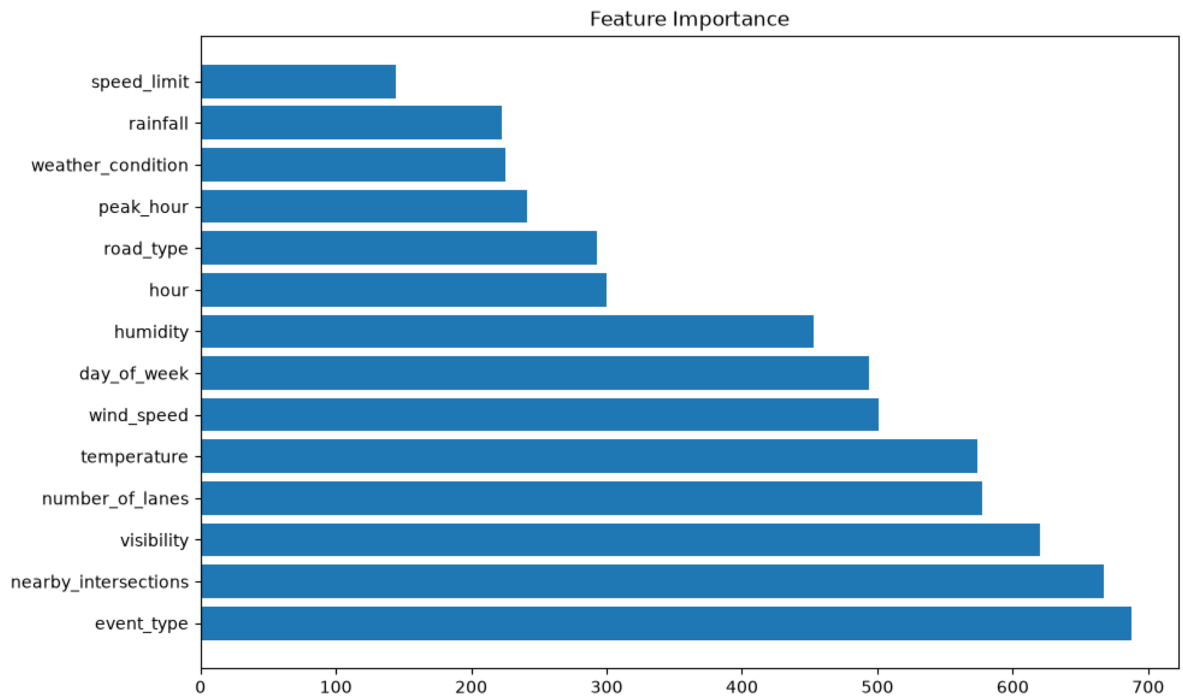

Fig3:Feature Importance

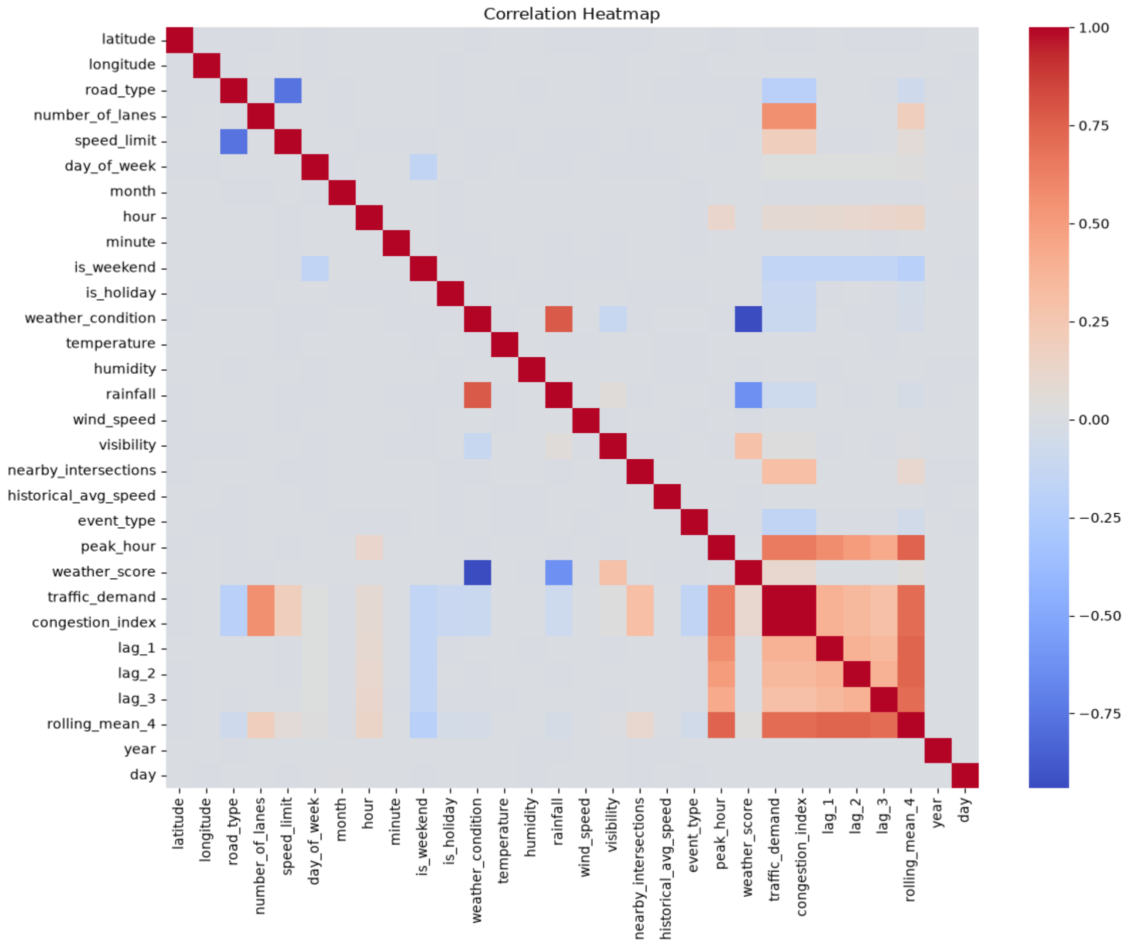

Fig4:Heatmap

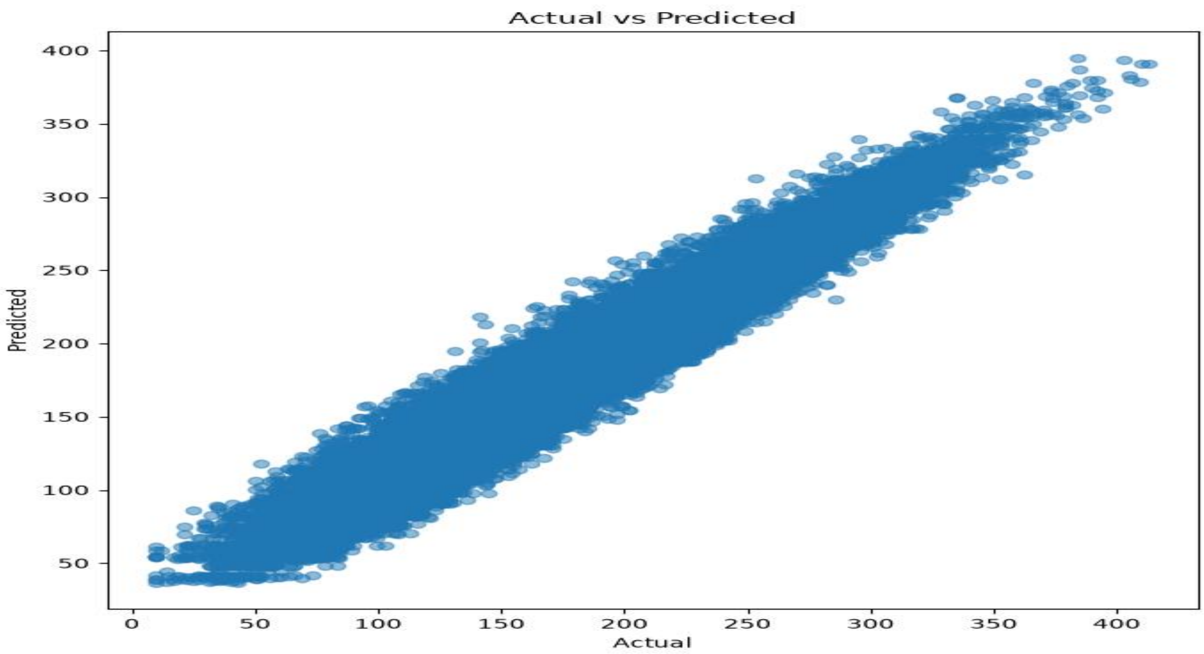

Fig5:Actual vs Predicted

Web Application: